

Distributed Database Benchmark

CockroachDB vs YugabyteDB

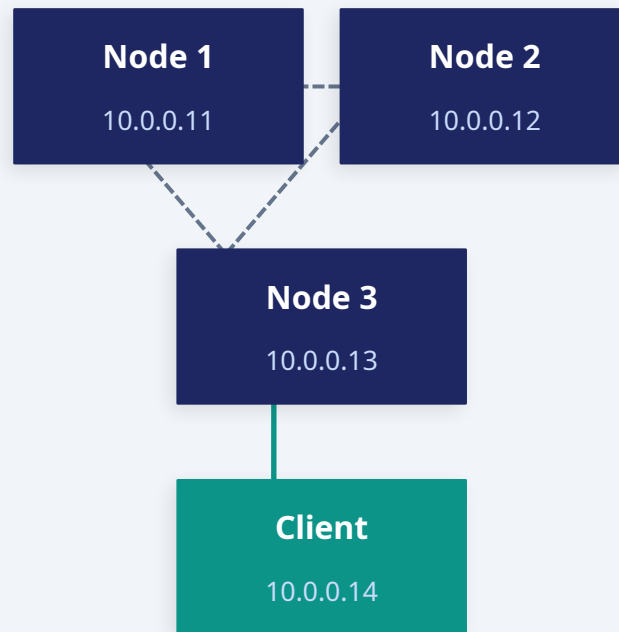
Network Partition Chaos Testing on AWS EC2

CockroachDB v24.1

YugabyteDB 2025.2

Test Architecture

Setting	Value
Cloud	AWS EC2
Instance Type	t3.medium
OS	Ubuntu 22.04
Storage	≥ 20 GB gp3
Network	Same VPC
Key Pair	Shared



DB Cluster (3 nodes) ↔ Benchmark Client

CockroachDB — Cluster Setup

1 Start each node (example: Node 1)

```
cockroach start --insecure \  
  --store=node1-data \  
  --listen-addr=10.0.0.11:26257 \  
  --http-addr=10.0.0.11:8080 \  
  --join=10.0.0.11:26257,10.0.0.12:26257,10.0.0.13:26257 \  
  --cache=1GB --max-sql-memory=1GB --background
```

2 Initialize the cluster (from client)

```
cockroach init --insecure --host=10.0.0.11:26257
```

3 Create database & load data

```
cockroach sql --insecure --host=10.0.0.11 -e "CREATE DATABASE benchmark;"  
pgbench -i -s 10 -h 10.0.0.11 -p 26257 -U root -d benchmark --init-steps=dtgpf
```

YugabyteDB — Cluster Setup

1 Start Node 1 (bootstrap)

```
./bin/yugabyted start \  
  --advertise_address=10.0.0.11 \  
  --cloud_location=aws.us-east-1.us-east-1a
```

2 Join remaining nodes

```
./bin/yugabyted start --advertise_address=10.0.0.12 \  
  --join=10.0.0.11 --cloud_location=aws.us-east-1.us-east-1b  
./bin/yugabyted start --advertise_address=10.0.0.13 \  
  --join=10.0.0.11 --cloud_location=aws.us-east-1.us-east-1c
```

3 Create database & load data

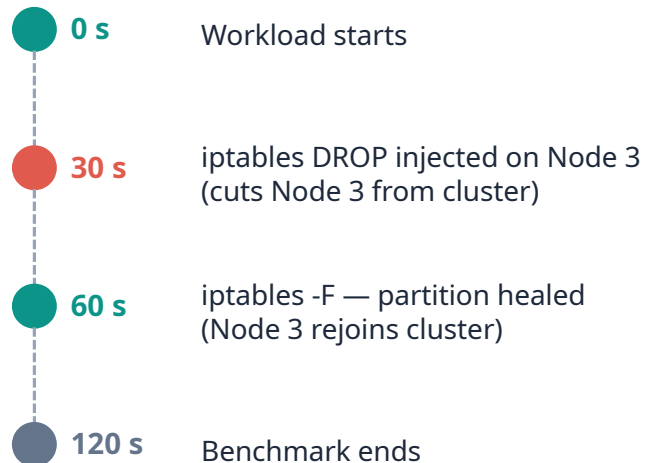
```
./bin/ysqlsh -h 10.0.0.11 -c "CREATE DATABASE benchmark;"  
pgbench -i -s 10 -h 10.0.0.11 -p 5433 -U yugabyte -d benchmark --init-steps=dtgpf
```

Benchmark Methodology

pgbench Configuration

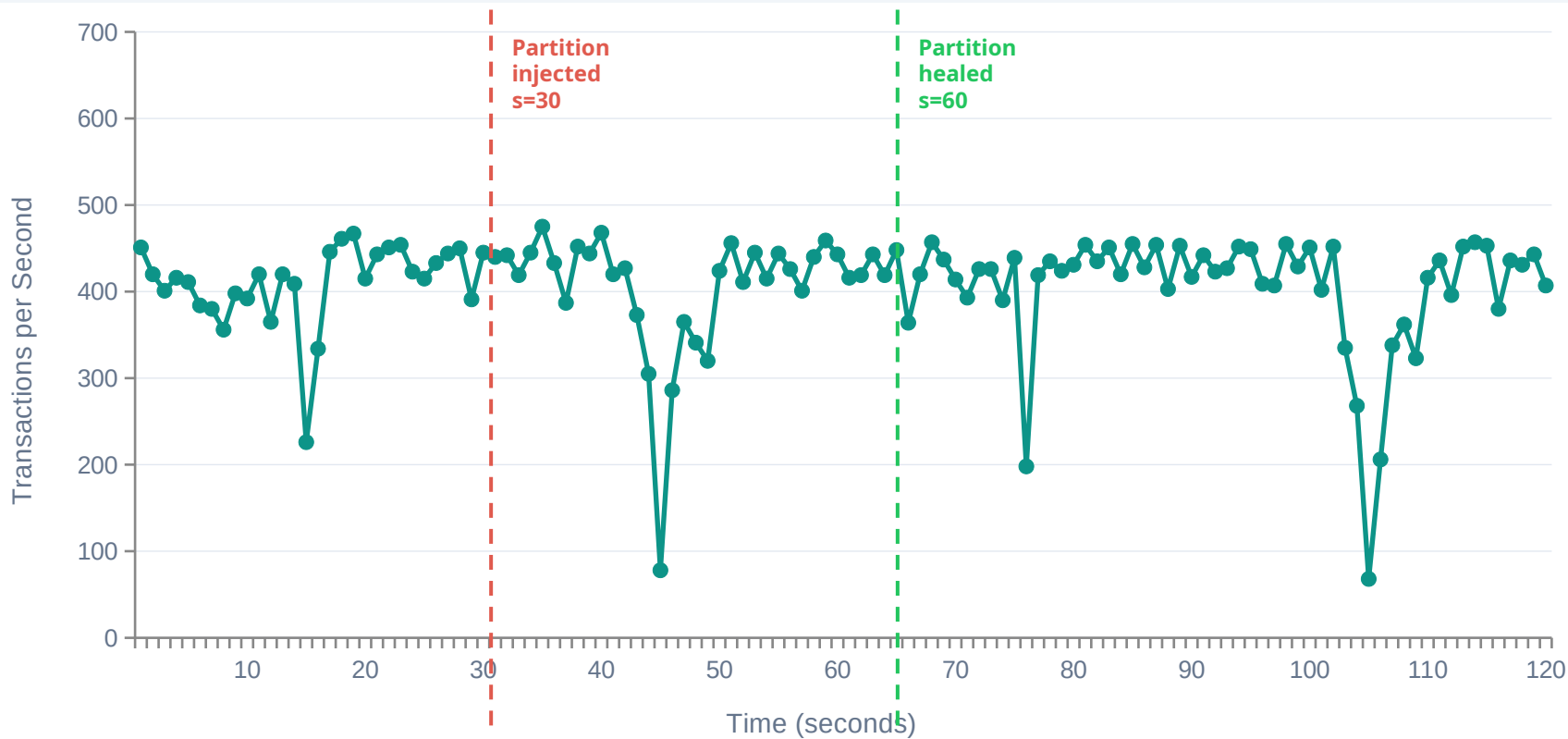
Clients (-c)	8
Threads (-j)	4
Duration (-T)	120 s
Scale factor (-s)	10 (1M rows)
Isolation	SERIALIZABLE
Progress interval	1 s

Network Partition Timeline

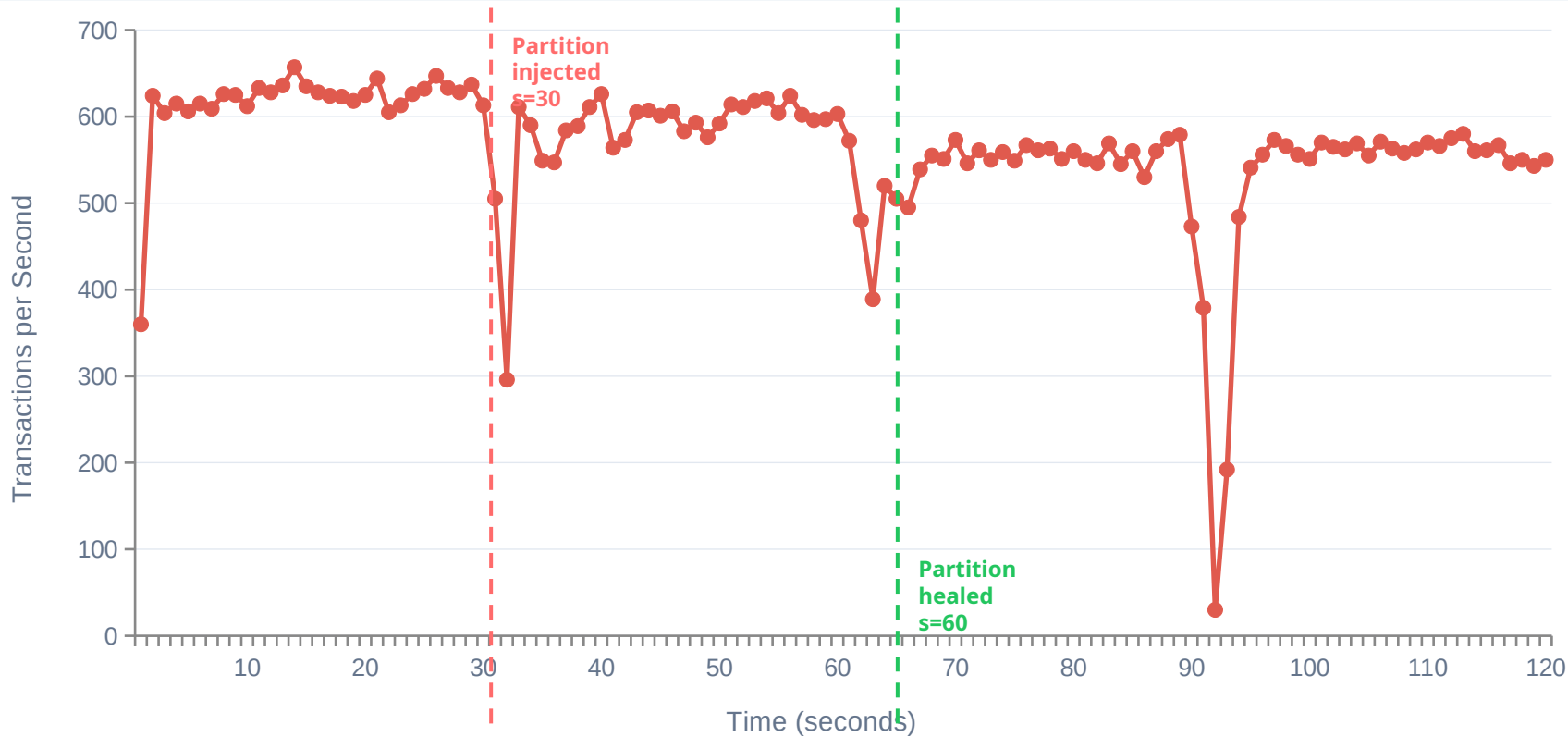


Transaction: SELECT + UPDATE on pgbench_accounts (custom serializable_txn.sql)

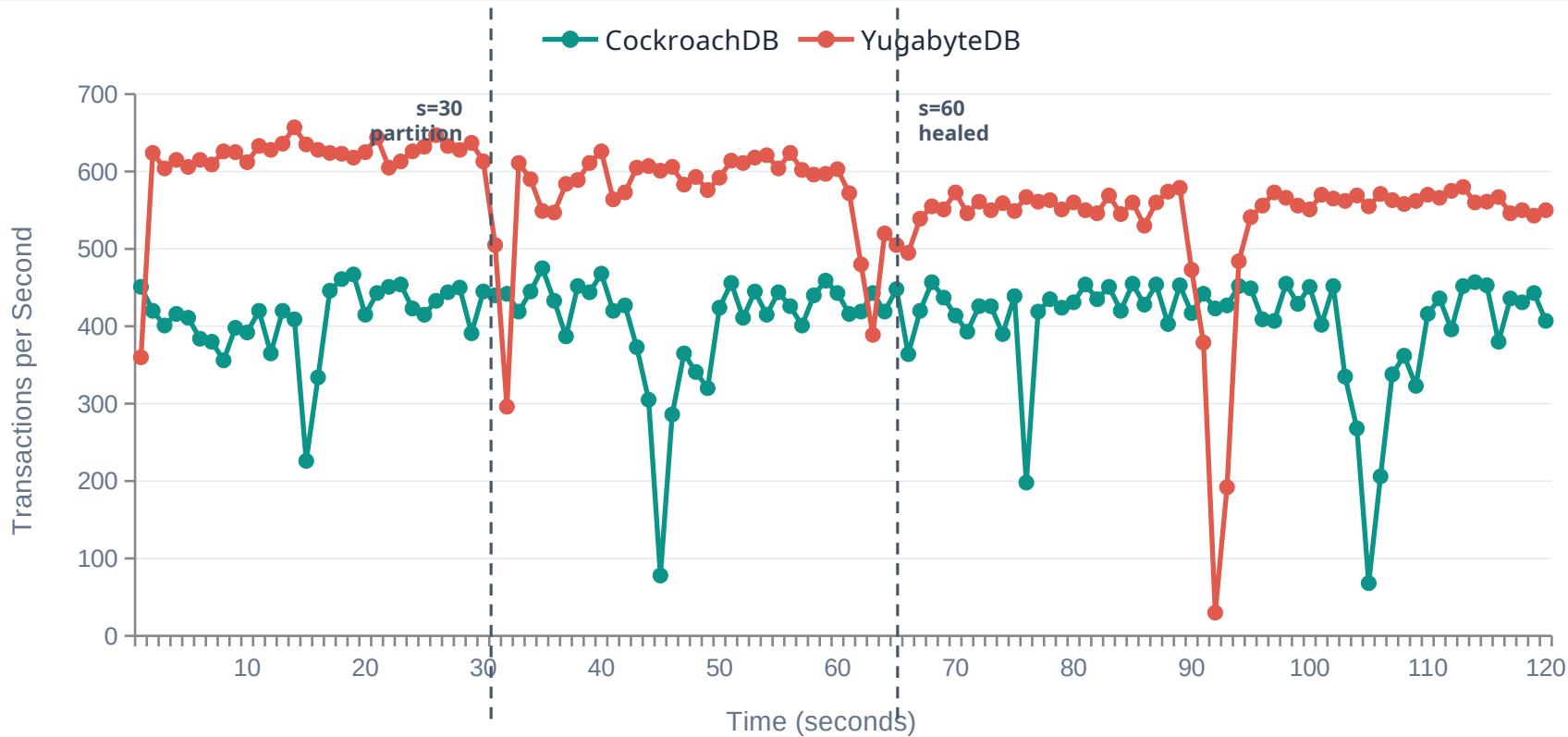
CockroachDB — TPS During Network Partition Test



YugabyteDB — TPS During Network Partition Test



TPS Comparison — Both Databases



Benchmark Results — Summary

Metric	CockroachDB	YugabyteDB
Avg TPS (overall)	406.7	566.9
Avg Latency	19.7 ms	14.1 ms
Latency Std Dev	8.8 ms	8.5 ms
Total Transactions	48,805	67,890
Failed Transactions	3 (0.006%)	0 (0.000%)
Min TPS during partition	68 tps (s=105)	30 tps (s=92)
Recovery time (initial partition s=30)	~15 s	~3 s
Recovery time (post-heal s=60)	~5 s	~3 s

Conclusions



YugabyteDB delivered 39% higher throughput (567 vs 407 TPS) and 28% lower latency (14.1 vs 19.7 ms) under the same workload.



YugabyteDB recovered from the initial partition in ~3 s. CockroachDB took ~15 s — Raft leader re-election over real WAN RTT adds measurable delay.



CockroachDB completed all 30 seconds of partition with 0 transaction failures while in the degraded window — strong consistency guarantee held.



YugabyteDB dropped to 30 TPS at second 92 (post-heal disruption), suggesting additional Raft re-balancing after the partition cleared.



Both systems maintained cluster integrity throughout the test: no data loss, no split-brain, consensus-safe behavior confirmed.